

ASSIGNMENT - 8

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BATCH-01

TASK-01:

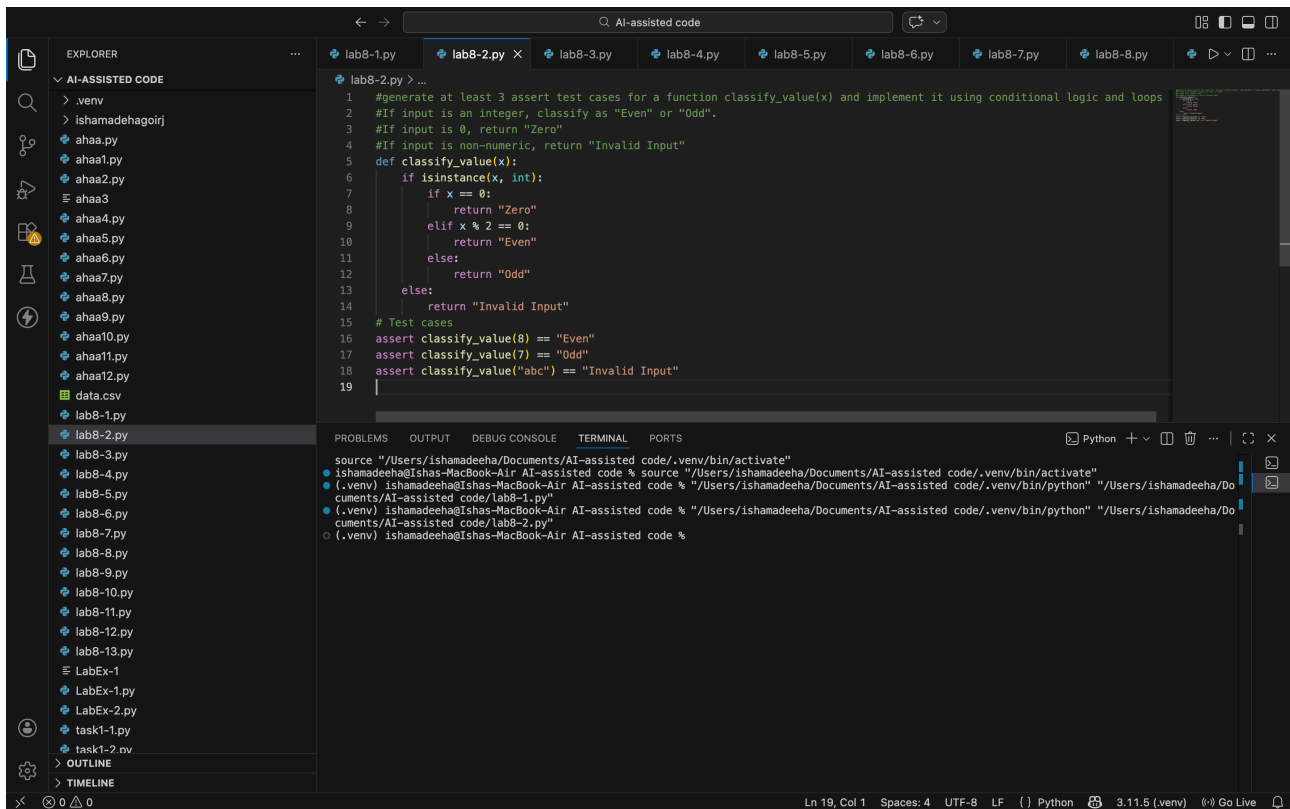
```
1 #generate at least 3 assert test cases for a function is_valid_username(username)
2 #Username length must be between 5 and 15 characters
3 #Must contain only alphabets and digits
4 #Must not start with a digit.
5 #No spaces allowed
6 def is_valid_username(username):
7     if len(username) < 5 or len(username) > 15:
8         return False
9     if not username.isalnum():
10        return False
11    if username[0].isdigit():
12        return False
13    if ' ' in username:
14        return False
15    return True
16 # Test cases
17 assert is_valid_username("User123") == True
18 assert is_valid_username("12User") == False
19 assert is_valid_username("Us er") == False
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
source "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/activate"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % source "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/activate"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % " /Users/ishamadeeha/Documents/AI-assisted code/lab8-1.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

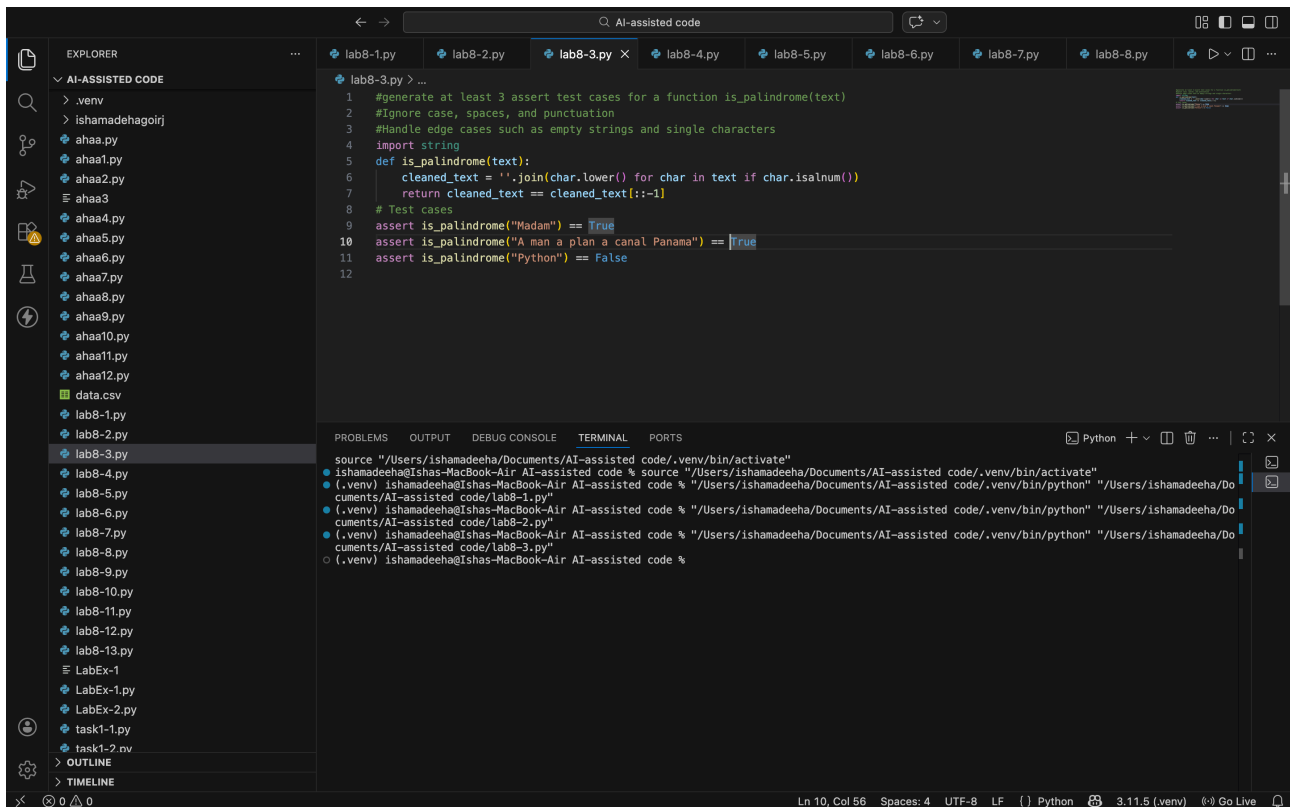
Ln 19, Col 43 Spaces: 4 UTF-8 LF Python 3.11.5 (.venv) Go Live

TASK - 02:



```
1 #generate at least 3 assert test cases for a function classify_value(x) and implement it using conditional logic and loops
2 #If input is an integer, classify as "Even" or "Odd".
3 #If input is 0, return "Zero"
4 #If input is non-numeric, return "Invalid Input"
5 def classify_value(x):
6     if isinstance(x, int):
7         if x == 0:
8             return "Zero"
9         elif x % 2 == 0:
10            return "Even"
11        else:
12            return "Odd"
13    else:
14        return "Invalid Input"
15 # Test cases
16 assert classify_value(8) == "Even"
17 assert classify_value(7) == "Odd"
18 assert classify_value("abc") == "Invalid Input"
19
```

TASK - 03:



```
1 #generate at least 3 assert test cases for a function is_palindrome(text)
2 #Ignore case, spaces, and punctuation
3 #Handle edge cases such as empty strings and single characters
4 import string
5 def is_palindrome(text):
6     cleaned_text = ''.join(char.lower() for char in text if char.isalnum())
7     return cleaned_text == cleaned_text[::-1]
8 # Test cases
9 assert is_palindrome("Madam") == True
10 assert is_palindrome("A man a plan a canal Panama") == True
11 assert is_palindrome("Python") == False
12
```

TASK-04:

The screenshot shows the VS Code editor interface. The Explorer panel on the left displays a file tree under 'AI-ASSISTED CODE' with files named 'ahaa1.py' through 'ahaa12.py', 'data.csv', and 'lab8-1.py' through 'lab8-13.py'. The main editor window is open to 'lab8-5.py', which contains the following Python code:

```
1 #generate a code to check a number is a perfect number (sum of divisors = number)
2 def is_perfect_number(n):
3     if n < 1:
4         return False
5     divisors_sum = sum(i for i in range(1, n) if n % i == 0)
6     return divisors_sum == n
7 # Test cases
8 assert is_perfect_number(6) == True # 1 + 2 + 3 = 6
9 assert is_perfect_number(10) == False
10 assert is_perfect_number(1) == False
```

The bottom panel shows the TERMINAL output with the following text:

```
source "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/activate"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % source "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/activate"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-1.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-2.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-3.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-4.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-5.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK - 05:

The screenshot shows the VS Code editor interface. The Explorer panel on the left displays a file tree under 'AI-ASSISTED CODE' with files named 'ahaa1.py' through 'ahaa12.py', 'data.csv', and 'lab8-1.py' through 'lab8-13.py'. The main editor window is open to 'lab8-5.py', which contains the following Python code:

```
1 #generate a code to check a number is a perfect number (sum of divisors = number)
2 def is_perfect_number(n):
3     if n < 1:
4         return False
5     divisors_sum = sum(i for i in range(1, n) if n % i == 0)
6     return divisors_sum == n
7 # Test cases
8 assert is_perfect_number(6) == True # 1 + 2 + 3 = 6
9 assert is_perfect_number(10) == False
10 assert is_perfect_number(1) == False
```

The bottom panel shows the TERMINAL output with the following text:

```
source "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/activate"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % source "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/activate"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-1.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-2.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-3.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-4.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Do
cuments/AI-assisted code/lab8-5.py"
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK -06:

The screenshot shows a VS Code editor with a file explorer on the left and a code editor in the center. The file explorer shows a project named 'AI-ASSISTED CODE' with a folder 'venv' and a file 'ishamadehagoirj'. The code editor shows a Python script named 'lab8-6.py' with the following code:

```
1 #generate a python code to Check if a number is abundant (sum of divisors > number)
2 def is_abundant_number(n):
3     if n < 1:
4         return False
5     divisors_sum = sum(i for i in range(1, n) if n % i == 0)
6     return divisors_sum > n
7 # Test cases
8 assert is_abundant_number(12) == True # 1 + 2 + 3 + 4 + 6 = 16 > 12
9 assert is_abundant_number(15) == False # 1 + 3 + 5 = 9 < 15
10 assert is_abundant_number(1) == False
```

The terminal at the bottom shows the output of the script, which is empty, indicating that the script is running successfully.

TASK - 07:

The screenshot shows a VS Code editor with a file explorer on the left and a code editor in the center. The file explorer shows a project named 'AI-ASSISTED CODE' with a folder 'venv' and a file 'ishamadehagoirj'. The code editor shows a Python script named 'lab8-7.py' with the following code:

```
1 #generate a python code to Check if a number is deficient (sum of divisors < number)
2 def is_deficient_number(n):
3     if n < 1:
4         return False
5     divisors_sum = sum(i for i in range(1, n) if n % i == 0)
6     return divisors_sum < n
7 # Test cases
8 assert is_deficient_number(8) == True # 1 + 2 + 4 = 7 < 8
9 assert is_deficient_number(12) == False # 1 + 2 + 3 + 4 + 6 = 16 > 12
10 assert is_deficient_number(1) == True # No divisors, sum = 0 < 1
```

The terminal at the bottom shows the output of the script, which is empty, indicating that the script is running successfully.

TASK - 08:

The screenshot shows a VS Code editor with a file explorer on the left and a code editor in the center. The file explorer shows a project named 'AI-ASSISTED CODE' with a subdirectory 'venv' and a file 'ishamadehagairj'. The code editor shows the implementation of a LeapYearChecker function and its test cases. The function is defined as follows:

```
1 #Write a function LeapYearChecker and validate its implementation using 10 pytest test cases
2 def LeapYearChecker(year):
3     if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
4         return True
5     else:
6         return False
7 # Test cases
8 def test_LeapYearChecker():
9     assert LeapYearChecker(2020) == True # Divisible by 4 and not
10    assert LeapYearChecker(1900) == False # Divisible by 100 but not 400
11    assert LeapYearChecker(2000) == True # Divisible by 400
12    assert LeapYearChecker(2021) == False # Not divisible by 4
13 def test_LeapYearChecker_additional():
14    assert LeapYearChecker(2400) == True # Divisible by 400
15    assert LeapYearChecker(1800) == False # Divisible by 100 but not
16    assert LeapYearChecker(1996) == True # Divisible by 4 and not
17    assert LeapYearChecker(2100) == False # Divisible by 100 but not 400
18    assert LeapYearChecker(2004) == True # Divisible by 4 and not
19    assert LeapYearChecker(2022) == False # Not divisible by 4
```

The terminal output shows the execution of the tests:

```
platform darwin -- Python 3.11.5, pytest-9.0.2, pluggy-1.6.0
rootdir: /Users/ishamadeeha/Documents/AI-assisted code
collected 2 items

lab8-8.py ..

===== 2 passed in 0.01s =====
```

TASK - 09:

The screenshot shows a VS Code editor with a file explorer on the left and a code editor in the center. The file explorer shows a project named 'AI-ASSISTED CODE' with a subdirectory 'venv' and a file 'ishamadehagairj'. The code editor shows the implementation of a SumOfDigits function and its test cases. The function is defined as follows:

```
1 #Write a function SumOfDigits and validate its implementation using 7 pytest test cases.
2 def SumOfDigits(n):
3     if n < 0:
4         return None # Return None for negative numbers
5     return sum(int(digit) for digit in str(n))
6 # Test cases
7 def test_SumOfDigits():
8     assert SumOfDigits(123) == 6 # 1 + 2 + 3 = 6
9     assert SumOfDigits(0) == 0
10    assert SumOfDigits(999) == 27 # 9 + 9 + 9 = 27
11    assert SumOfDigits(456) == 15 # 4 + 5 + 6 = 15
12    assert SumOfDigits(-123) == None # Negative number
13    assert SumOfDigits(1001) == 2 # 1 + 0 + 0 + 1 = 2
14    assert SumOfDigits(7) == 7 # Single digit number
```

The terminal output shows the execution of the tests:

```
platform darwin -- Python 3.11.5, pytest-9.0.2, pluggy-1.6.0
rootdir: /Users/ishamadeeha/Documents/AI-assisted code
collected 2 items

lab8-8.py ..

===== 2 passed in 0.01s =====

platform darwin -- Python 3.11.5, pytest-9.0.2, pluggy-1.6.0
rootdir: /Users/ishamadeeha/Documents/AI-assisted code
collected 1 item

lab8-9.py .

===== 1 passed in 0.00s =====
```

TASK -10:

The screenshot shows a VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files 'ahaa1.py' through 'ahaa12.py' and 'lab8-1.py' through 'lab8-11.py'. The main editor displays the file 'lab8-10.py' with the following code:

```
1 #Write a function SortNumbers (implement bubble sort) and validate its implementation using 25 pytest test cases
2 def SortNumbers(arr):
3     n = len(arr)
4     for i in range(n):
5         for j in range(0, n-i-1):
6             if arr[j] > arr[j+1]:
7                 arr[j], arr[j+1] = arr[j+1], arr[j]
8     return arr
9 # Test cases
10 def test_SortNumbers():
11     assert SortNumbers([5, 2, 9, 1, 5, 6]) == [1, 2, 5, 5, 6, 9]
12     assert SortNumbers([1]) == [1]
13     assert SortNumbers([3]) == [3]
14     assert SortNumbers([3, 2]) == [2, 3]
15     assert SortNumbers([1, 2, 3]) == [1, 2, 3]
16     assert SortNumbers([3, 2, 1]) == [1, 2, 3]
17     assert SortNumbers([5, -1, 0]) == [-1, 0, 5]
18     assert SortNumbers([-3, -1, -2]) == [-3, -2, -1]
19     assert SortNumbers([10, 9, 8]) == [8, 9, 10]
20     assert SortNumbers([1.5, 0.5, -0.5]) == [-0.5, 0.5, 1.5]
21     assert SortNumbers([1, 2, 2, 3]) == [1, 2, 2, 3]
22     assert SortNumbers([5, 4, 3, 2, 1]) == [1, 2, 3, 4, 5]
23     assert SortNumbers([1, 1, 1]) == [1, 1, 1]
24     assert SortNumbers([0, 0, 0]) == [0, 0, 0]
25     assert SortNumbers([1, 0, 1]) == [0, 1, 1]
26     assert SortNumbers([2, 3, 1]) == [1, 2, 3]
```

The terminal at the bottom shows the execution of the tests:

```
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m pytest lab8-9.py
platform darwin -- Python 3.11.5, pytest-9.0.2, pluggy-1.6.0
rootdir: /Users/ishamadeeha/Documents/AI-assisted code
collected 1 item

lab8-9.py . [100%]

(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m pytest lab8-10.py
platform darwin -- Python 3.11.5, pytest-9.0.2, pluggy-1.6.0
rootdir: /Users/ishamadeeha/Documents/AI-assisted code
collected 1 item

lab8-10.py . [100%]

(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK - 11:

The screenshot shows a VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files 'ahaa1.py' through 'ahaa12.py' and 'lab8-1.py' through 'lab8-11.py'. The main editor displays the file 'lab8-11.py' with the following code:

```
1 #Write a function ReverseString and validate its implementation using 5 unittest test cases
2 def ReverseString(s):
3     return s[::-1]
4 import unittest
5 class TestReverseString(unittest.TestCase):
6     def test_ReverseString(self):
7         self.assertEqual(ReverseString("hello"), "olleh")
8         self.assertEqual(ReverseString("Python"), "nohtyP")
9     def test_ReverseString_edge_cases(self):
10         self.assertEqual(ReverseString(""), "")
11         self.assertEqual(ReverseString("a"), "a")
12         self.assertEqual(ReverseString("12345"), "54321")
13 if __name__ == '__main__':
14     unittest.main()
```

The terminal at the bottom shows the execution of the tests:

```
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m pytest lab8-9.py
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m pytest lab8-10.py
platform darwin -- Python 3.11.5, pytest-9.0.2, pluggy-1.6.0
rootdir: /Users/ishamadeeha/Documents/AI-assisted code
collected 1 item

lab8-10.py . [100%]

(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m pytest lab8-11.py
..
Ran 2 tests in 0.000s

OK
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK -12:

```
lab8-12.py > TestAnagramChecker > test_AnagramChecker_edge_cases
1 #Write a function AnagramChecker and validate its implementation using 10 unittest test cases.
2 def AnagramChecker(str1, str2):
3     return sorted(str1.replace(" ", "").lower()) == sorted(str2.replace(" ", "").lower())
4 import unittest
5 class TestAnagramChecker(unittest.TestCase):
6     def test_AnagramChecker(self):
7         self.assertTrue(AnagramChecker("listen", "silent"))
8         self.assertTrue(AnagramChecker("Triangle", "Integral"))
9         self.assertFalse(AnagramChecker("apple", "pabble"))
10    def test_AnagramChecker_edge_cases(self):
11        self.assertTrue(AnagramChecker("Dormitory", "Dirty Room"))
12        self.assertFalse(AnagramChecker("Hello", "World"))
13        self.assertTrue(AnagramChecker("A gentleman", "Elegant man"))
14        self.assertTrue(AnagramChecker("Clint Eastwood", "Old West Action"))
15        self.assertTrue(AnagramChecker("School master", "The classroom"))
16        self.assertTrue(AnagramChecker("Astronomer", "Moon starrer"))
17        self.assertTrue(AnagramChecker("Conversation", "Voices rant on"))
18 if __name__ == '__main__':
19     unittest.main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Documents/AI-assisted code/lab8-12.py"

Traceback (most recent call last):
File "/Users/ishamadeeha/Documents/AI-assisted code/lab8-12.py", line 16, in test_AnagramChecker_edge_cases
self.assertFalse(AnagramChecker("Astronomer", "Moon starrer"))
AssertionError: True is not false

Ran 2 tests in 0.001s

FAILED (failures=1)
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Documents/AI-assisted code/lab8-12.py"

..

Ran 2 tests in 0.000s

OK
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code %

TASK -13:

```
lab8-13.py > ...
2 def ArmstrongChecker(n):
3     return armstrong_sum == n
4
5 import unittest
6 class TestArmstrongChecker(unittest.TestCase):
7     def test_ArmstrongChecker(self):
8         self.assertTrue(ArmstrongChecker(153)) # 1^3 + 5^3 + 3^3 = 153
9         self.assertTrue(ArmstrongChecker(370)) # 3^3 + 7^3 + 0^3 = 370
10        self.assertTrue(ArmstrongChecker(9474)) # 9^4 + 4^4 + 7^4 + 4^4 = 9474
11    def test_ArmstrongChecker_non_armstrong(self):
12        self.assertFalse(ArmstrongChecker(123)) # Not an Armstrong number
13        self.assertFalse(ArmstrongChecker(10)) # Not an Armstrong number
14    def test_ArmstrongChecker_edge_cases(self):
15        self.assertTrue(ArmstrongChecker(0)) # 0 is an Armstrong number
16        self.assertTrue(ArmstrongChecker(1)) # 1 is an Armstrong number
17
18 if __name__ == '__main__':
19     unittest.main()
20
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Documents/AI-assisted code/lab8-12.py"

Ran 2 tests in 0.001s

FAILED (failures=1)
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code % "/Users/ishamadeeha/Documents/AI-assisted code/.venv/bin/python" "/Users/ishamadeeha/Documents/AI-assisted code/lab8-12.py"

..

Ran 2 tests in 0.000s

OK
(.venv) ishamadeeha@Ishas-MacBook-Air AI-assisted code %